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Integration of geospatial techniques and soil loss modelling for soil conservation in a drought prone river basin of eastern India

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Abstract

Soil erosion is a major environmental challenge in India, where millions of tonnes of fertile soil are lost annually, adversely affecting agricultural productivity and ecosystem sustainability. Reliable spatial assessment of erosion risk is therefore essential for prioritizing conservation interventions. In parts of eastern India, characterized by intense monsoonal rainfall, undulating topography, and dynamic land-use patterns, soil erosion remains a critical concern. Within this region, parts of the Mahanadi River basin have been identified as highly susceptible to soil erosion. Jonk River Basin, being a part of Mahanadi basin, has been recognized by the National Bureau of Soil Survey and Land Use Planning as a highly vulnerable area for soil conservation; however, detailed spatial assessment of soil erosion at basin and sub-basin scales remains largely unexplored. This study presents the novel contribution for spatial assessment of soil erosion in the Jonk River Basin using the Revised Universal Soil Loss Equation (RUSLE) integrated with Geographic Information Systems (GIS), remote sensing, and statistical computing in the R programming language. Rainfall, soil properties, topography, land-use/land-cover, and conservation practice datasets were integrated to derive RUSLE factors and generate basin-wide soil loss estimates. In addition, a factor influence analysis was performed in R to quantify the relative contribution of individual erosion-controlling parameters. The results reveal that 4.81% of the basin area falls under moderate to very high erosion risk classes, with soil loss ranging from 55 to 137.5 t ha^{-1} yr^{-1}. Sub-basin level analysis identifies SB-8 as the most erosion-prone region, primarily due to its moderately to highly dissected plateau terrain with hills and valleys that enhance runoff and sediment transport, whereas SB-5 experiences relatively low erosion risk. Model validation using field-observed erosion sites and Receiver Operating Characteristic (ROC) analysis demonstrates strong predictive performance, with an Area Under the Curve (AUC) value of 0.97, indicating very high model accuracy. The Random Forest-based importance analysis revealed the dominant role of slope length and steepness (LS) and conservation practice (P) factors in controlling erosion dynamics within the basin. The findings also highlight significant spatial heterogeneity in erosion processes and critical management gaps, including exposed slopes lacking protective measures and the prevalence of water-intensive paddy monocropping in drought-prone conditions. The integration of RUSLE with GIS, remote sensing, and

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the R environment provides a reproducible and scalable framework for soil erosion assessment. The study therefore, provides a scientific basis for targeted soil and water conservation planning and sustainable watershed management in erosion-prone regions.

Keywords River basin, Soil erosion rate, Soil loss, RUSLE, Modelling, ROC curve

1 Introduction

Soil erosion is a critical environmental challenge worldwide, and India faces particularly severe impacts due to its diverse physiography, intense monsoonal rainfall, and extensive dependence on agriculture. Estimates suggest that India loses approximately 5334 million tonnes of soil annually, corresponding to an average of 16.4 tonnes per hectare per year, largely as a result of water erosion processes [33]. As a fundamental natural resource, soil supports agricultural production, sustains biodiversity, and maintains the health and functioning of ecosystems [2]. Soil erosion reduces agricultural productivity, diminishes soil fertility, and contributes to sedimentation in rivers and reservoirs, therefore threatening both food security and environmental sustainability [63, 103]. Additionally, changes in land use and land cover such as deforestation, urbanization, and agricultural expansion, have intensified climate variability. Intensive agricultural practices have further degraded soil quality, particularly during dry periods, and have contributed to alterations in rainfall and temperature patterns [3]. In Eastern India, recent studies reveal several erosion hotspots across diverse regions including Chota Nagpur Plateau [89], Murredu watershed, Deccan Plateau [118], and eastern Godavari sub-basins [76]. To address these challenges, identification of soil erosion affected zones and modelling the erosion rate are essential for conservation planning. Among soil erosion models, the Universal Soil Loss Equation (USLE) was first developed in the United States as an empirical model to predict long-term average annual soil loss from sheet and rill erosion [139]. However, as research progressed, several limitations of USLE were recognized, as it did not adequately account for temporal variation in rainfall patterns, soil processes, and land cover changes, and lacked flexibility in handling complex topographic and management conditions [105].

The Revised Universal Soil Loss Equation (RUSLE) was developed in the 1990s to address these shortcomings. RUSLE retained the empirical structure of USLE but introduced significant improvements [9, 105], such as improvements in soil erodibility (K) estimation for different soil textures, refinements in LS-factor equations to better represent complex topographies, and methods for deriving C and P factors using remote sensing and GIS integration. By integrating these diverse components, RUSLE provides a comprehensive means of quantifying the spatial distribution of erosion risk across landscapes [120]. Over time, the RUSLE model has been widely applied across diverse environmental settings worldwide [96], ranging from semi-arid watersheds in India [14] to tropical forest regions of the Amazon [71] and Africa [6], demonstrating its broad applicability as a global tool for soil erosion assessment. Numerous recent studies have successfully applied RUSLE to address soil erosion in different regions of India, such as the Igra watershed of the Chhota Nagpur Plateau [89],the southern Western Ghats [1, 130–132],the Murredu watershed, Telangana [118]; the Middle Godavari sub-basin [76],the Nethravathi basin, southern India [43],the Silabati River catchment [88],the Dikrong River catchment, north-eastern India [27],the Kallar watershed, Tamil Nadu [34],and

the Upper Krishna River Basin [97], highlighting the suitability of the RUSLE model for assessing soil erosion under diverse geomorphic and climatic conditions. The integration of Geographic Information Systems (GIS) and remote sensing (RS) with RUSLE has made it widely recognized as a standard model for soil erosion risk assessment and conservation planning.

The Mahanadi River Basin is one of the most significant river basins of eastern India and plays a crucial role in hydrology, agriculture, and socio-economic development of the region [61, 62]. The basin is characterized by diverse physiographic features including plateau regions, forested uplands, alluvial plains, and a well-developed deltaic region in coastal Odisha, with high erosion susceptibility particularly in the upper plateau and forested uplands due to steep slopes, lateritic soils, and intense monsoon rainfall [31]. Major tributaries such as the Seonath, Hasdeo, Mand, Ib, Ong, and Jonk rivers contribute significantly to the drainage network and hydrological dynamics of the basin, with the upper catchment (Chhattisgarh-Odisha plateau region) identified as the most vulnerable to soil erosion [28, 126]. Previous investigations in the Mahanadi River basin, Eastern India, have extensively applied geospatial techniques to examine watershed characteristics, hydrological behaviour, and soil erosion dynamics. Several studies have employed the Revised Universal Soil Loss Equation (RUSLE) integrated with GIS to estimate soil erosion risk and sediment dynamics across different parts of the basin. Basin-scale modelling approaches have been used to simulate annual soil loss and sediment transport processes, demonstrating the applicability of integrated erosion–sediment models and high-performance computational techniques for large watershed systems [28]. The influence of climate variability on soil erosion has also been investigated, indicating a decline in mean erosion rates between 1981 and 2019, while simultaneously reporting an increase in the spatial extent of erosion-prone areas [31]. Similarly, RUSLE-based assessments in sub-catchments such as the Hirakud catchment have estimated average annual soil loss of approximately 29 t ha^{-1} yr^{-1} and identified priority zones for soil conservation to reduce reservoir sedimentation [126]. Additional studies within the basin have focused on sediment yield estimation in the upper Mahanadi region and erosion dynamics in the southern delta using DEM and land cover-based approaches [79, 110].

The Jonk River, a key right-bank tributary of the upper Mahanadi, forms part of the most vulnerable upper plateau zone of the Mahanadi catchment and therefore represents a priority area for RUSLE–GIS-based modelling to support soil conservation and protect downstream water resources. As per the soil data provided by the National Bureau of Soil Survey and Land Use Planning (NBSS&LUP), the Jonk River Basin is experiencing soil erosion ranging from severe to very severe levels. The Jonk River Basin has historically experienced prolonged drought conditions. In 2002, parts of the basin were affected by drought and hence came under the DPAP (Drought Prone Area Development Program) scheme. During extended droughts, the reduction in vegetation cover, soil moisture, and biological activity typically leads to the deterioration of soil structure and an increase in the exposure of bare soil surfaces. These conditions increase the vulnerability of the landscape to erosional processes once precipitation resumes. Whenever intense or erratic rainfall occurs in such areas, there is a significant increase in surface runoff due to the hardened or crusted nature of the soil formed during the dry phase. This can result in accelerated soil erosion, particularly in sloping areas or regions lacking

adequate ground cover. Moreover, areas previously stabilized by sparse vegetation may experience gully formation or expansion due to the sudden increase in runoff volume and velocity. Thus, the Jonk River flows from drought-affected regions causing severe soil erosion. The eroded sediments are transported to the downstream stretches of the river, where their deposition reduces the channel's capacity to convey excess water and sediment during the monsoon season. Consequently, identifying and mapping the spatial extent of soil erosion within the basin is essential for watershed management and effective planning.

Previous studies have examined many aspects of the Jonk River Basin. For example, geospatial approaches have been used to delineate groundwater potential zones for the drought-prone area through multi-criteria decision-making techniques [10]. Morphometric and geomorphological analyses have also been carried out to understand the spatial characteristics and evolution of landforms within the basin [66, 119]. Furthermore, watershed-scale studies in the basin have investigated hydrological responses to land-use and land-cover changes as well as the role of conservation structures in watershed management [5, 48, 65]. These studies provide insights into hydrology, geomorphology, and watershed processes in the region. However, despite being classified under severe to very severe soil erosion susceptibility, the Jonk River Basin, located in the Chhattisgarh–Odisha region as an upper tributary of the Mahanadi system, has received no scientific attention with respect to spatially explicit soil erosion assessment. This gap is significant, as erosion processes in such sub-basins strongly influence sediment yield, reservoir siltation, land degradation, and downstream river morphology within the Mahanadi system.

In this context, the present study makes a novel contribution by providing a detailed, spatially explicit assessment of soil erosion at the sub-basin scale, focusing exclusively on the hydrologically delineated Jonk River Basin using the RUSLE model integrated with GIS and remote sensing techniques. The study not only estimates the soil loss rate but also examines sub-basin-level variability including spatial prioritization, identifies erosion-prone hotspots, and validates the model results using field observations and statistical analysis. By addressing the absence of erosion-focused studies in this tributary of the Mahanadi basin, the research contributes new insights into the spatial dynamics of soil erosion and provides a scientific basis for watershed-scale soil conservation planning and sustainable land management in this erosion-prone region. This study fills the critical research gap by integrating topography, land use, soil characteristics, rainfall erosivity, and conservation practices. It provides a high-resolution spatial understanding of erosion patterns that has not been previously documented for this basin and contributes insights for targeted soil conservation planning and sustainable watershed management within the larger Mahanadi River system. In addition to estimating soil loss using the RUSLE model, understanding the relative influence of individual erosion-controlling factors is essential for identifying the dominant drivers of soil erosion. Quantifying the contribution of rainfall erosivity, soil erodibility, topography, land cover, and conservation practices helps determine which environmental conditions most strongly control erosion processes within a basin. Therefore, a statistical analysis was carried out to examine the relationship between RUSLE factors and soil loss and to evaluate their importance on erosion dynamics in the Jonk River Basin. The study provides detailed factor-wise analysis and Random Forest-based relative importance evaluation of RUSLE

factors. The factor influence analysis was performed using the R programming language environment, which enabled systematic computation of correlations and relative contributions of individual RUSLE parameters. R provides a reproducible platform for integrating statistical modelling with geospatial datasets, allowing efficient handling of large spatial layers and automation of analytical workflows. Through this approach, the study was able to quantitatively assess the dominance of specific factors controlling soil erosion. The use of R not only enhances analytical precision but also ensures transparency and reproducibility, thereby strengthening the overall methodological framework and contributing to the advancement of GIS-based environmental modelling.

In this study, the main objectives are: (1) to quantify the spatial distribution of soil erosion risk across the basin using the RUSLE model integrated with geospatial techniques; (2) to identify and prioritize erosion-prone hotspots at the sub-basin level; (3) to evaluate the relative influence of RUSLE factors (R, K, LS, C, and P) on soil erosion in the basin; (4) to validate the model outputs using ROC analysis and field-based observations; and (5) to examine the key factors contributing to soil erosion and suggest targeted conservation measures for effective watershed management.

2 Study area

The Jonk River Basin is a part of the drainage system of the Mahanadi River, which is one of India's major east-flowing river systems. The river covers two districts, Nuapada (880.42 km^2) and Bargarh (86.13 km^2) of Odisha state (966.55 km^2), and three districts, Mahasamund (1612.78 km^2), Gariyaband (34.38 km^2), and Baloda Bazar (840.38 km^2) of Chhattisgarh state (2487.53 km^2). The basin is located between 20°28′13.78″ N to 21°43′50.40″ N latitudes and 82°19′6.24″ E to 82°51′21.42″ E longitudes (Fig. 1). The

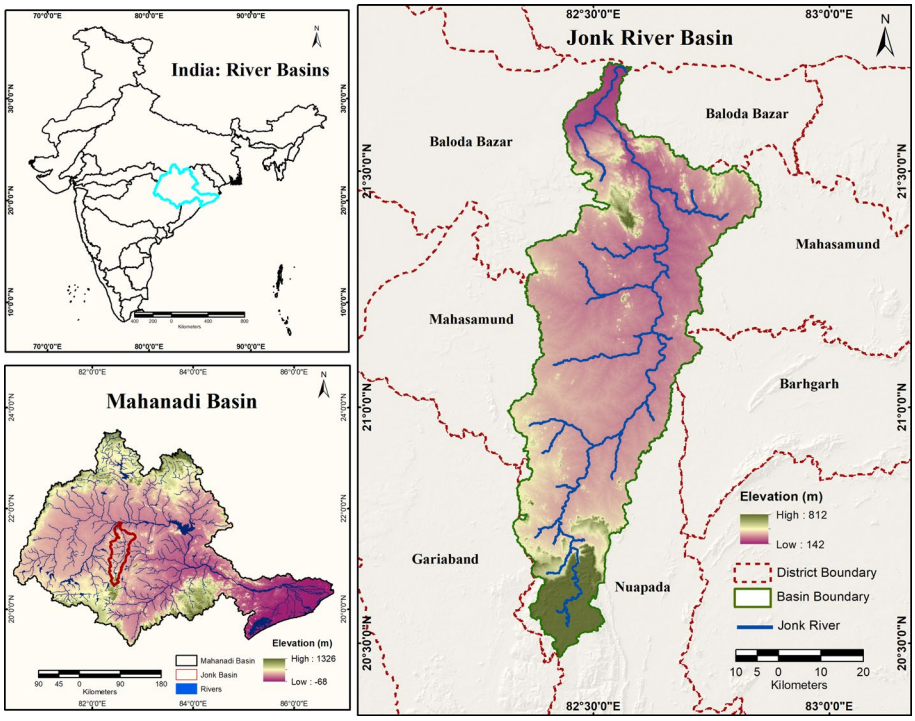


Fig. 1 Locational extent of Jonk river basin. *Source*: Prepared using SRTM DEM and Survey of India (SOI) administrative boundary

total catchment area of the river is about 3454.08 km^2 [119]. The basin is sub-divided into eight sub-basins based on its eight tributaries of seventh order, in which sub-basin (SB)-1 is the largest and sub-basin (SB)-7 is the smallest catchment. The southern region of the basin near the Sunabeda plateau, the northern and north-eastern region, the south-western region, and the central hills have steep to very steep slopes, which can be visible in SB-8, SB-3, and SB-2. The rest of the basin has gentle to very gentle slope, which reflects flat terrain with low relief features. The study area experiences a tropical monsoon type climate where mean annual temperature ranges from a maximum of 47 °C in the dry season (May) to a minimum of 6 °C in the winter season (January). Average humidity varies from 40 to 95%, with maximum humidity being felt in August and minimum in April and May. During the monsoon season, storms or depressions formed in the Bay of Bengal cross the east coast moving westward and causing heavy rain in the region. Average annual rainfall in the region varies from 1221 to 1257 mm, and 80% of the annual rainfall is received during the S-W monsoon, and August is the rainiest month of the season.

3 Database and methodology

3.1 Preparation of base layers

Table 1 and Fig. 2 show the datasets used and the comprehensive methodological framework adopted for estimating soil loss using the RUSLE model integrated with GIS techniques. SRTM (Shuttle Radar Topography Mission; 30 m resolution) DEM (Digital Elevation Model) was used for the delineation of basin boundary using ArcGIS Pro spatial analyst extension and ArcHydro Tool. The Survey of India toposheets (1:50,000 scale) were used to prepare stream segments, which were manually digitized according to their hierarchical orders following Strahler's scheme of stream ordering [123] in ArcGIS Pro software, to ensure accurate delineation of drainage features and maintain spatial precision of the drainage network. Manual digitization from toposheets is widely used in geomorphological and watershed studies to accurately represent stream segments and avoid errors that may arise from automated extraction methods [58, 61, 62]. Sub-basin boundaries were delineated using pour points with sixth-order stream threshold in ArcGIS software using the ArcSWAT extension. SRTM DEM was used to extract slope in SAGA GIS (7.8.2). Previous studies have shown that 30 m SRTM DEM provides reliable topographic information with consistent spatial resolution, global coverage, and

Table 1 Database of thematic layers used and their source of extraction

Dataset	Source
Soil	FAO (Food and Agricultural Organisation)
Drainage network	Digitized from Survey of India Toposheets 64K/6, 64K/10, 64K/14, 64K/7, 64K/11, 64K/15, 64K/8, 64K/12, 64L/5, 64L/9, 64L/6, 64L/10, 64L/7, and 64L/11 (1:50,000 scale) in Arc GIS (10.5)
Satellite Imagery (2022)	Landsat (9, OLI/TRIS); USGS Earth explorer portal extracted in *Google Earth Engine (GEE)*
Digital Elevation Model	SRTM (Shuttle Radar Topography Mission; 30 m resolution)
Slope	Extracted from DEM in SAGA GIS (7.8.2)
Rainfall data (1984–2022)	Extracted from NASA POWER in R Studio using (API) Application Programming Interface (0.5°×0.625° resolution)
Basin and sub-basin boundary	Extracted from SRTM DEM in Arc SWAT
NDVI (Normalized Difference Vegetation Index)	Extracted from Landsat (9, OLI/TRIS); USGS Earth explorer portal in *Google Earth Engine (GEE)*

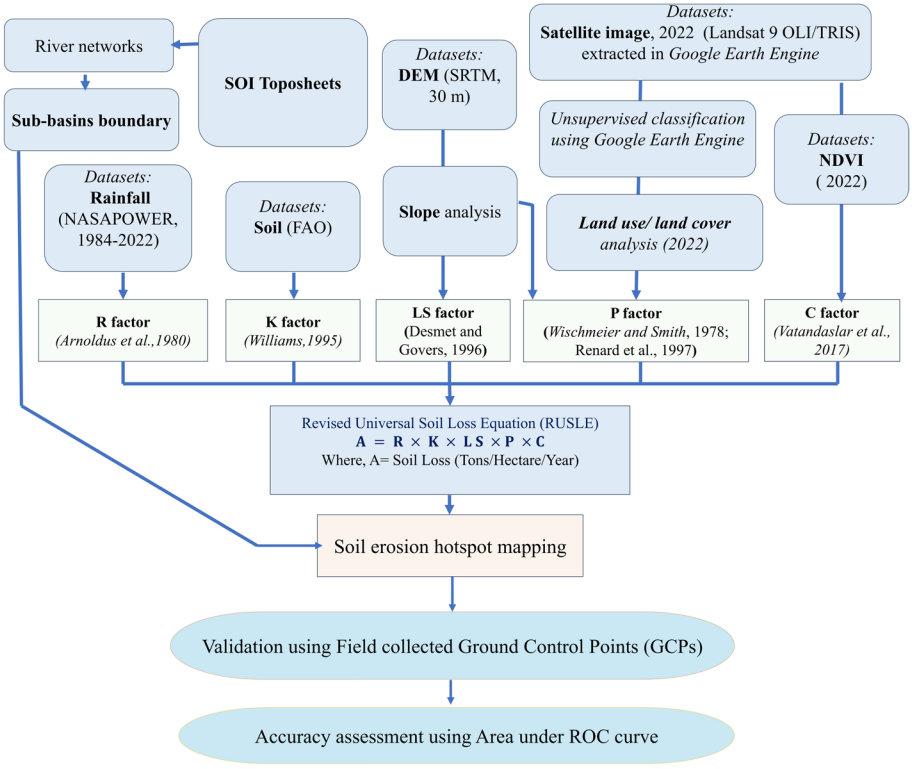


Fig. 2 Methodological flowchart for soil loss estimation

acceptable vertical accuracy for assessment of soil erosion due to slope steepness [104, 124, 147], while SAGA GIS offers advanced terrain analysis tools for deriving slope and hydrological parameters from DEM data required for RUSLE-based studies [104, 124].

Rainfall data were obtained from the NASA POWER (Prediction of Worldwide Energy Resources) database using an Application Programming Interface (API) in RStudio (version 4.3.0) for the period 1984–2022 (39 years). The long-term rainfall record was selected because the rainfall erosivity factor (R-factor) in the Revised Universal Soil Loss Equation (RUSLE) represents the average annual erosive power of rainfall, which requires multi-decadal precipitation records to capture inter-annual variability and long-term rainfall trends. Therefore, a 39-year rainfall record was used to obtain a reliable mean rainfall value required in the empirical equation proposed by Arnoldus [9]. The API-based data retrieval provides automated, reproducible, and efficient extraction of large long-term datasets. Kriging interpolation was used to generate the rainfall map in ArcGIS because it considers spatial autocorrelation among rainfall observations, producing statistically optimized and unbiased estimates of rainfall distribution across the study area [20, 49].

The soil database was obtained from the Food and Agriculture Organization (FAO) global soil database. Key attributes including soil texture (sand, silt, clay fractions) and organic matter content were used to compute the K-factor following standard formulations [105, 140]. The FAO soil dataset was selected because it provides harmonized and spatially consistent soil property information at regional and global scales, which is particularly useful in data-scarce regions lacking detailed field-based soil surveys. Several recent studies have also utilized FAO soil datasets for estimating the soil erodibility

factor and implementing the RUSLE model in watershed-scale soil erosion assessments [22, 46].

Continuous raster datasets, including the FAO-derived soil erodibility (K) layer and rainfall run-off erosivity (R) layer were resampled to 30 m using the bilinear interpolation method during projection and spatial harmonization. Bilinear interpolation was selected because it is widely recommended for continuous raster surfaces, where output cell values are estimated using weighted averages of neighbouring pixels, thereby preserving gradual spatial transitions and minimizing abrupt discontinuities during downscaling and resampling procedures.

The Land use/land cover (LULC) map of 2022 was used for the estimation of the support practice factor (P-factor) in the RUSLE model. The P-factor represents current land management and conservation practices, which depend on the present land use conditions of the study area [12, 81, 121]. Therefore, recent LULC data (2022) were used to accurately represent existing land management practices affecting soil erosion. Recent studies increasingly rely on cloud-computing platforms such as Google Earth Engine (GEE) for land use and land cover (LULC) classification because it enables rapid processing of large satellite datasets and integration of machine-learning algorithms for accurate mapping. The platform provides access to multi-temporal satellite imagery (e.g., Sentinel-2,Landsat data) and built-in classifiers such as Random Forest, Support Vector Machine, and CART for supervised classification, and K-means for unsupervised classification, making it a widely adopted tool for contemporary LULC analysis and environmental monitoring [57, 115, 150]. The LULC map for 2022 was prepared using Landsat-9 OLI/TIRS (Collection-2 Tier-1, TOA reflectance) imagery obtained from USGS (United States Geological Survey) Earth Explorer portal in the Google Earth Engine (GEE) platform (see the Supplement, Figure S1). Images acquired between 01 February and 01 March 2022 with <5% cloud cover were selected, clipped to the Jonk River Basin, and a median composite was generated in the GEE code editor. Unsupervised K-means clustering was applied to classify the satellite imagery, as it allows automatic grouping of spectrally similar pixels without requiring prior training data [53, 70]. This approach is particularly suitable for heterogeneous landscapes where reliable ground reference information is limited [42, 144], such as the Jonk basin. The resulting spectral clusters were subsequently interpreted and merged based on visual interpretation and ancillary information to produce seven final LULC classes, following established practices in RUSLE-GIS erosion studies across Indian river basins [4, 89]: Water Bodies (WB), Built-up (BU), Cropland (CL), Dense Forest (DF), Open Forest (OF), Barren Land (BL), and Sand Bar (SB). Although unsupervised classification may introduce ambiguity particularly in distinguishing spectrally similar classes such as open forest, cropland, and also in detecting settlements, these limitations were addressed through systematic post-processing where clusters were iteratively evaluated and corrected based on field knowledge and visual interpretation. Post-processing of the LULC classification was carried out in ERDAS IMAGINE (2014) to improve accuracy. The classified output was refined through recoding using high-resolution Google Earth imagery, with particular emphasis on the accurate delineation of settlement areas for enhancing classification accuracy and reducing misclassification errors prior to analysis. The majority filter is applied to remove isolated pixels.

Additionally, 210 field verification points collected using a mobile GPS during field surveys, along with high-resolution Google Earth imagery, were used as reference data to validate the classification and improve the reliability of the final 2022 LULC map (see the Supplement; Figures S2, S3). The accuracy of the classified LULC map was evaluated using a confusion matrix based on 210 reference samples. The results indicate a high level of classification accuracy, with an overall accuracy of 95.71% and a kappa coefficient of 0.94, which shows strong agreement between the classified map and reference data (see the Supplement; Table S1).

Normalized Difference Vegetation Index (NDVI) was calculated for the year 2022 to represent the present vegetation cover conditions of the watershed. NDVI derived from satellite imagery is widely used to estimate the RUSLE cover-management factor because it effectively represents spatial vegetation density and soil protection conditions [11, 37, 41, 73, 129, 142, 145]. NDVI was also derived and processed in GEE using median Landsat 9 imagery (01 February and 01 March 2022). NDVI is calculated from the near-infrared (NIR) and red spectral bands to represent vegetation density. For Landsat-9, Band 5 (NIR) and Band 4 (Red) were used to compute NDVI using the standard formula given below.

$$\mathrm{NDVI} = (\mathrm{NIR} + \mathrm{Red})/(\mathrm{NIR} - \mathrm{Red}) \tag{1}$$

$$\text{For Landsat 9,} \quad \mathrm{NDVI} = (\mathrm{Band\,5} + \mathrm{Band\,4})/(\mathrm{Band\,5} - \mathrm{Band\,4}) \tag{2}$$

3.1.1 Rainfall-runoff erosivity factor (R)

The rainfall erosivity factor (R) represents the influence of precipitation on soil erosion by accounting for both rainfall amount and intensity. In the RUSLE model, it is commonly estimated using the EI_{30} index, which combines the total storm kinetic energy with the maximum 30-min rainfall intensity [105]. Originally introduced by Wischmeier and Smith [140], the EI_{30} index is calculated as one-hundredth of the product of storm kinetic energy and its peak 30-min rainfall intensity and is widely used to represent the erosive potential of rainfall. However, its application in India, particularly at basin or district scales, is limited due to the lack of consistent high-temporal-resolution rainfall data. Most available meteorological datasets provide only daily or monthly rainfall totals, which are insufficient for direct EI_{30} estimation. Recent studies have also emphasized that the scarcity of pluviographic records and high-resolution rainfall observations introduces uncertainty in rainfall erosivity estimation and soil erosion modelling [72].

To address this limitation, several studies have developed empirical approaches to estimate rainfall erosivity using long-term monthly or annual rainfall data through regression relationships and rainfall-based indices such as the Modified Fournier Index (MFI) and precipitation concentration indices [9, 15, 21, 50, 85, 90, 91, 93, 146, 151]. As rainfall erosivity varies considerably across different climatic regimes, many studies have further developed region-specific empirical equations that adapt these relationships to local rainfall characteristics, enabling reliable estimation of the R-factor in data-scarce regions [21, 25, 44]. In this study, the R-factor was estimated using an empirical formulation based on monthly precipitation, as proposed by Arnoldus [9].

$$\mathrm{R} = \sum_{i}^{12} 1.735 \times 10 \left[1.5 \times \log_{10} \left(\frac{pi^2}{p} \right) - 0.08188 \right] \tag{3}$$

where R = rainfall erosivity factor (MJ mm ha^{-1} h^{-1} yr^{-1}); Pi = monthly rainfall (mm); P = annual rainfall (mm). With the above equation, the R-factor was calculated using the Raster Calculator function in ArcGIS using average monthly and annual rainfall data (1984–2022).

3.1.2 Soil erodibility factor (K)

Soil erodibility represents the inherent susceptibility of soil to detachment and transport by erosive forces such as rainfall impact and surface runoff. It is primarily influenced by soil properties including texture, structure, organic matter content, permeability, and aggregate stability. The soil erodibility factor (K) reflects the resistance of soil to erosion and indicates the rate of soil loss under standard rainfall and runoff conditions [105, 140]. Soil characteristics such as particle size distribution, soil structure, porosity, and organic carbon content significantly affect the magnitude of the K factor [113]. Recent studies have further emphasized the role of soil physical and chemical properties in controlling soil erodibility and its spatial variability under different environmental conditions [22, 69, 90, 91, 93]. Several empirical equations have been proposed to estimate the soil erodibility factor, among which the formulation developed by Williams [138] is widely applied in erosion modelling frameworks. In the present study, the soil erodibility factor (K) was estimated using the equation proposed by Williams [138].

$$\mathrm{K} = F_{csand} \times F_{cl-si} \times F_{hisand} \times F_{orgc} \tag{4}$$

$$F_{csand} = 0.2 + 0.3 \times exp \left[-0.256 \times m_{sand} \left(1 - \frac{m_{silt}}{100} \right) \right] \tag{5}$$

$$F_{cl-si} = \left[\frac{m_{silt}}{(m_{cl} + m_{silt})} \right]^{0.3} \tag{6}$$

$$F_{orgc} = 1 - \frac{0.25\, orgC}{orgC + exp\left[3.72 - 2.95 \times orgC\right]} \tag{7}$$

$$F_{hisand} = 1 - \frac{0.7 \left(1 - \frac{m_{sand}}{100}\right)}{\left(1 - \frac{m_{sand}}{100}\right) + exp\left[-5.51 + 22.9 \left(1 - \frac{m_{sand}}{100}\right)\right]} \tag{8}$$

where orgC = organic carbon content (%) of topsoil; m_{sand} = sand content (%) of topsoil; m_{silt} = silt content (%) of topsoil; and m_{cl} = clay content (%) of topsoil.

Using these soil parameters, the K factor was calculated in Microsoft Excel (365) using these equations. Then, the K values were assigned to the respective soil classes in the prepared layer in ArcGIS Pro.

3.1.3 Slope length and steepness factor (LS)

The slope length and steepness factor (LS) represents the effect of topography on soil erosion by integrating the influence of slope length (L) and slope steepness (S) [78, 83]. The slope length component reflects the accumulation of overland flow, where longer slopes allow runoff to concentrate and increase its erosive energy. Similarly, an increase

in slope steepness accelerates runoff velocity and enhances the detachment and transport capacity of soil particles, thereby increasing erosion rates. Consequently, areas characterized by longer and steeper slopes are generally more vulnerable to soil erosion [8, 149].

In GIS-based erosion modelling, the LS factor is commonly derived from Digital Elevation Model (DEM) data using slope gradient and flow accumulation algorithms. Recent studies have highlighted the importance of DEM-based topographic analysis for accurately estimating the LS factor and its role in controlling the spatial variability of soil erosion [7, 13, 86]. In the present study, the LS factor was estimated within the SAGA GIS environment using the LS-factor module. This module employs a raster-based formulation consistent with the approach proposed by Desmet and Govers and incorporates a multiple flow direction algorithm to ensure realistic representation of flow distribution in complex terrain. The adoption of the Desmet and Govers [35] method is particularly suitable for grid-based DEM analysis, as it effectively captures spatial variability and improves the estimation of flow accumulation in heterogeneous landscapes,therefore, it is widely recognized over the unit stream power approach given by Moore and Burch [82], which was developed for continuous and uniform slopes and thus leads to oversimplification of the LS factor in complex terrain [80, 133, 148]. The LS factor was computed using the following equation [35]:

$$\mathrm{LS} = \left[\mathrm{FA} \times \frac{cell\ size}{22.13}\right]^{0.4} \times \left[\frac{\mathrm{Sin\,Slope}}{0.0896}\right]^{1.3} \tag{9}$$

where *FA* represents the flow accumulation, *cell size* is the spatial resolution of the DEM, and θ is the slope angle in radians. The exponent values were adopted following standard RUSLE-based applications.

In this formulation, flow accumulation derived within SAGA GIS represents the upslope contributing area, while slope gradient is calculated from the DEM (30 m) and incorporated through the sine function by the software itself. The use of the multiple flow direction algorithm in SAGA GIS allows for improved representation of runoff dispersion compared to single-flow direction methods, which is particularly important for heterogeneous and complex landscapes such as the Sunabeda plateau. Moreover, SAGA performs calculations faster than traditional GIS software and facilitates the derivation of flow accumulation directly from Digital Elevation Models (DEMs), which improves the efficiency of terrain modelling [38, 75, 87, 102, 114].

3.1.4 Cover and management factor (C)

The C factor is calculated to analyse the effect of cropping and management practices (soil-disturbing activities, vegetation, crop sequence, productivity level, soil cover, and subsurface biomass) on erodibility. The C factor quantifies the relative soil loss from land managed under particular cropping systems and practices, compared to the soil loss from bare land subjected to continuous clean tillage without any conservation measures [140]. It has a close linkage with land use types and measures the reduction factor in soil loss. The value of the C factor depends mainly on vegetation health, type, stage of crop development, and cover percentage. There is a linear correlation between NDVI and the C factor. Multiple equations for different regions have been developed by many scholars

using NDVI [32, 37, 134]. In this study, the C factor was calculated using the equation proposed by Vatandaşlar and Yavuz [134].

$$C = 0.431 - [0.805 \times (NDVI)] \tag{10}$$

Recent literature indicates that this equation has been widely utilized in regional and basin-scale soil erosion assessments across diverse environmental settings, including Himalayan watersheds, semi-arid basins, and tropical landscapes [29, 127]. Moreover, comprehensive review studies on C-factor estimation methods have identified this formulation as one of the commonly adopted NDVI-based empirical approaches in contemporary RUSLE applications because of its operational simplicity, compatibility with satellite-derived vegetation indices, and suitability for large-area spatial modelling [92, 143]. These studies collectively support the applicability of the adopted equation within GIS and remote sensing-based soil erosion assessment frameworks.

3.1.5 Conservation practice factor (P)

The conservation practice factor (P) represents the effect of soil conservation and land management practices in reducing soil erosion [36, 39, 141]. It reflects the impact of support practices such as terracing, contour farming, strip cropping, and afforestation that are implemented to minimize surface runoff and soil loss [99, 128]. The P factor therefore indicates the efficiency of these conservation measures in controlling erosion processes. Lower P-factor values correspond to areas where effective conservation practices are applied, while higher values indicate the absence or limited implementation of such measures [36, 106, 112]. The values of the P factor range from 0 to 1, where higher values are generally assigned to areas with no conservation practices, while lower values correspond to built-up land and cropped areas. Lower P-factor values indicate higher effectiveness of the applied conservation practices in reducing soil erosion [106, 140].

Different conservation practices give different results in protecting soil from erosion. Several scholars have proposed tables for estimating the P factor by integrating slope with conservation practices. For example, Renard et al. [106] calculated P values that relate slope percentage to practices such as contour farming, strip cropping, and terracing following USDA guidelines. Wischmeier and Smith [140] suggested P-factor values based on slope and land use, but the method was limited as it only accounts for agricultural land and assigns uniform values to non-agricultural areas regardless of land cover type. A limitation of this equation is that it does not account for conservation practices applied to either agricultural or non-agricultural land, and it considers only slope steepness without incorporating land use.

In this study, the P-factor map was generated by integrating both LULC and slope characteristics. A reference table (Table 2), compiled from existing literature, was used to assign P-factor values by considering both land use/conservation practices and slope steepness. First, the slope of the study basin (derived from DEM) was converted into percentage rise using the slope function of the Surface Spatial Analyst tool in ArcGIS. It was then reclassified into discrete classes using the Reclass function of the Spatial Analyst tool. Using the 'Combine' function of the Spatial Analyst tool, the reclassified slope map was combined with the land use/land cover map of the study basin. Using the editor tool, P values were assigned to each slope subclass of each LULC class as per Table 2.

Table 2 P value for LULC classes combined with slope steepness

Land use/land cover	Slope (%)	P factor value
Agriculture	0–5	0.1
	5–10	0.12
	10–20	0.14
	20–30	0.19
	30–50	0.25
	>50	0.7
Water	0–100	0
Urban land	0–100	0.05
Forest	0–5	0.03
	5–10	0.05
	10–20	0.1
	31–50	0.2
	>50	0.5
Rangeland	0–5	0.1
	5–10	0.13
	11–20	0.15
	21–30	0.2
	30–50	0.4
	>50	0.75
Wasteland	0–5	0.25
	5–10	0.35
	10–20	0.45
	20–30	0.55
	30–50	0.75
	>50	1

Source: Modified after Wischmeier and Smith [140] and Renard et al. [106]

3.2 Spatial identification of soil erosion hotspots and soil loss estimation

After deriving all five input factors, the extent of soil erosion was estimated using the Revised Universal Soil Loss Equation (RUSLE) model by multiplying these factors to obtain the average annual soil loss, based on the equation:

$$A = R \times K \times LS \times C \times P \tag{11}$$

where A = Average annual soil loss per unit area (t ha^{-1} yr^{-1}), R = Rainfall-runoff erosivity factor (MJ mm ha^{-1} h^{-1} yr^{-1}); K = Soil erodibility factor (t ha h ha^{-1} MJ^{-1} mm^{-1}); LS = Slope length and slope steepness factor; C = Cover and management factor; and P = Conservation practice factor.

3.3 Calculation of relative influence of factors on soil erosion estimates

3.3.1 Descriptive statistics and correlation-based association of RUSLE factors

To evaluate the relative influence of the RUSLE factors on soil erosion, a statistical analysis was performed using the R programming environment (version 4.5.2). Raster layers representing rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C), and conservation practice (P), along with the estimated soil loss layer (A), were imported into R using the 'terra' package [51]. Prior to analysis, all raster layers were harmonized to ensure identical spatial resolution, spatial extent, and coordinate reference system. This was achieved by projecting the raster layers to a common coordinate system and resampling them to match the grid structure of the soil loss raster. Subsequently, the aligned raster layers were combined into a raster stack and converted

into a tabular dataset by extracting pixel values while excluding missing values or no data value. The resulting dataset contained corresponding values of all RUSLE factors and soil loss for each pixel across the study area. A Pearson's correlation analysis was then performed to quantify the strength and direction of the linear relationship between each RUSLE factor and the estimated soil loss values [26, 77, 98]. To examine the statistical association between individual RUSLE factors and estimated soil loss, Pearson correlation analysis was performed using randomly sampled raster values. The Pearson correlation coefficient (r) was calculated between soil loss and each RUSLE factor (R, K, LS, C, and P) to evaluate the direction and strength of their linear relationship.

For visualization, a bar plot showing the correlation-based heat map of the RUSLE factors was generated using the ggplot2 package [137] to improve interpretability, and to distinguish positive and negative relationships.

3.3.2 Random Forest-based variable importance analysis

To obtain a statistically robust estimate of the relative contribution of individual RUSLE factors toward soil erosion variability, a Random Forest (RF) regression approach was employed. Random Forest is a non-parametric ensemble machine learning algorithm that constructs multiple decision trees and combines their predictions to improve model stability and predictive performance [24]. The method is widely used in environmental and geospatial modelling because it can effectively handle nonlinear relationships, complex interactions among variables, and multicollinearity issues.

Initially, all raster layers corresponding to the RUSLE factors, namely rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover-management (C), conservation practice (P), and estimated soil loss (A), were spatially aligned to ensure identical projection, resolution, and extent using the *terra* package in R. Subsequently, the aligned raster layers were stacked into a multi-layer raster dataset for further analysis.

To reduce computational complexity while preserving spatial variability, 10,000 random sample pixels were extracted from the raster stack using the random sampling function available in the *terra* package. Sampling was restricted to valid non-missing pixels only. The extracted dataset included values of all predictor variables (R, K, LS, C, and P) along with the corresponding soil loss values.

The Random Forest regression model was then implemented using the *randomForest* package in R [68], where soil loss was considered the dependent variable and the RUSLE factors were treated as predictor variables. The model was trained using 500 decision trees (*ntree* = 500) to ensure stable estimation of variable importance.

Variable importance was evaluated using the percentage increase in Mean Squared Error (%IncMSE), which measures the increase in prediction error when the values of a particular predictor are randomly permuted while all other variables are retained unchanged. Higher %IncMSE values indicate greater contribution of the corresponding variable to model prediction accuracy. To facilitate comparative interpretation, the obtained variable importance values were normalized into relative importance percentages using the following equation:

$$RI_i = \frac{|VI_i|}{\sum |VI_i|} \times 100 \tag{12}$$

where RI_i represents the relative importance percentage of the ith variable and VI_i denotes the Random Forest variable importance value (%IncMSE) of that variable.

3.4 Validation of the results

Validation of model outputs is a significant step to assess the reliability of predicted zones. In this study, the accuracy of the RUSLE-based soil erosion model in the Jonk River Basin was evaluated using three approaches: (1) comparison of predicted erosion areas with observed erosion-prone locations, (2) statistical validation using Receiver Operating Characteristic (ROC) curve analysis, and (3) comparison of estimated soil erosion rates with results from previous studies conducted in Indian river basins, including the Mahanadi River Basin, of which the study basin is a part. Ground-truthing was conducted to validate soil erosion conditions using field-based observations. Field verification points were assessed based on standard geomorphological indicators, including the presence of rills and gullies, sheet erosion features, sediment deposition zones, and exposed or disturbed topsoil. A site was classified as erosion-affected if one or more of these characteristics were observed. A total of 24 erosion points (ground control points—GCPs) were collected during field surveys using a mobile GPS device and used as validation samples (see the supplementary material; Table S2). These points were overlaid on the predicted soil erosion map and used to generate the ROC curve using the ArcSDM tool in ArcGIS. The ROC curve evaluates model performance by plotting the true positive rate (sensitivity) against the false positive rate (1 – specificity) at different threshold values; thus, it assesses the model's ability to correctly distinguish erosion-prone areas from non-erosion areas [84]. For ROC analysis, the predicted soil erosion map was reclassified into binary classes (using the Reclassify function in ArcGIS), which represent erosion-prone and non-erosion areas based on threshold values derived from erosion severity classes. Field-observed erosion locations were treated as "presence" (true condition), while randomly selected non-erosion locations from stable areas were considered as "absence" points. Using this dataset, true positives (correctly predicted erosion locations), false positives (non-erosion locations predicted as erosion), true negatives, and false negatives were systematically derived by the ArcSDM tool. The tool then used these values to construct the Receiver Operating Characteristic (ROC) curve and compute the Area Under the Curve (AUC). The Area Under the Curve (AUC), derived from the ROC curve, provides a quantitative measure of prediction accuracy, where values closer to 1 indicate higher model reliability, while values near 0.5 represent random prediction [22, 101]. ROC-based validation has been widely applied in recent soil erosion and RUSLE-based modelling studies to assess the predictive capability of erosion risk zones [19, 88].

4 Results

4.1 Rainfall characteristics and erosivity factor (R)

The mean annual rainfall across the Jonk River Basin ranges from 1553 mm in the north to 1803 mm in the southern regions (Fig. 3). This gradient demonstrates spatial heterogeneity in rainfall distribution, which has direct implications for erosivity. Figure 3 shows that the southern and southwestern zones, receiving the highest rainfall, are more susceptible to intense erosive forces due to heavy downpours, while the relatively drier northern stretches experience comparatively lower erosive energy. Rainfall erosivity (R

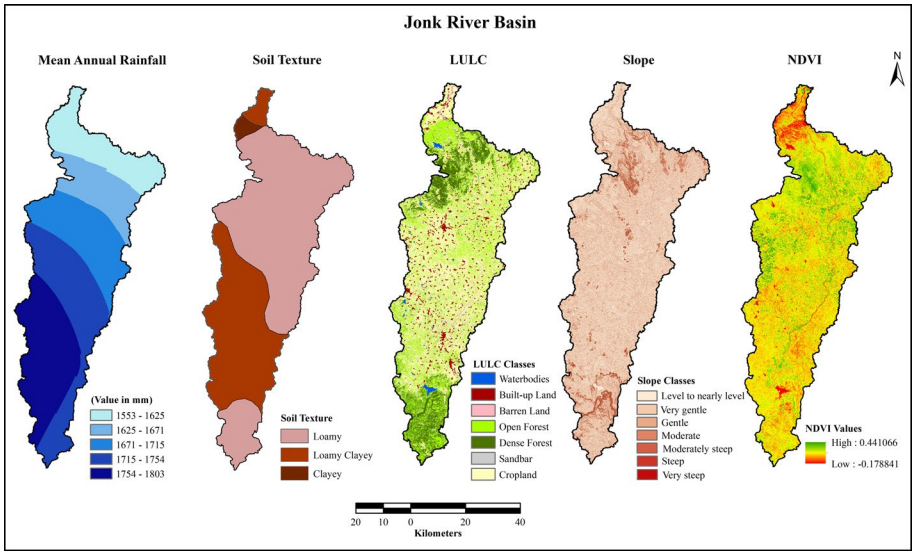


Fig. 3 Base layers used for the preparation of input layers of RUSLE modelling: mean annual rainfall; soil texture; land use/land cover (LULC); slope; and NDVI. *Source*: Prepared using NASA power rainfall data, FAO soil, Landsat 9 image (2022), and SRTM DEM

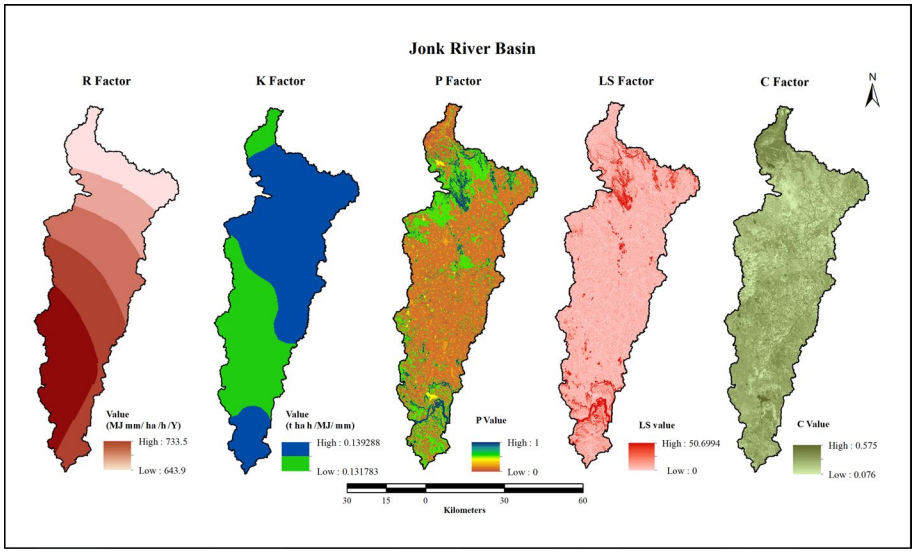


Fig. 4 Input parameters for soil loss estimation using RUSLE modelling: R factor; K factor; P factor; LS factor; and C factor. *Source*: Prepared using parameters extracted in Fig. 3

factor) is a critical driver of spatial variability in erosion risk, as higher precipitation intensity increases surface runoff, soil particle detachment, and sediment transport [59].

The rainfall erosivity (R) factor quantifies the impact of rainfall intensity and kinetic energy on soil detachment and transport [30]. Spatial variability within the Jonk River Basin shows that R-factor values range between 733.5 and 643.9 MJ mm ha^{-1} h^{-1} yr^{-1} (Fig. 4). Higher values of the R factor are concentrated in the southern and southwestern parts of the basin, reflecting regions of intense rainfall events, while relatively lower values are observed in the northern sub-basins. This gradient highlights the uneven rainfall distribution across the basin, where southern zones are more vulnerable to

storm-induced soil erosion. The R factor is therefore a critical determinant of erosion hotspots, as areas with both high rainfall intensity and susceptible soils exhibit higher erosion risk.

4.2 Soil texture and soil erodibility (K-factor)

Soil texture in the basin is dominated by loamy, loamy-clayey, and clayey soils, each influencing soil erodibility differently (Fig. 3). Loamy soils, which are well-drained and moderately cohesive, are moderately erodible due to their balance between sand, silt, and clay. Loamy-clayey soils, concentrated in the central basin, exhibit reduced infiltration capacity and greater susceptibility to runoff under intense rainfall, contributing to moderate-to-high erodibility values. Clayey soils, found largely in the southern basin, are generally more resistant to detachment due to their fine texture and cohesion; however, once detached, clay particles are easily transported by runoff. This variation in soil properties directly contributes to the K factor of RUSLE, representing the susceptibility of soils to erosion under standard conditions.

The soil erodibility (K) factor represents the susceptibility of soils to erosion, primarily controlled by properties such as soil texture, structure, organic matter, and permeability. Using FAO soil data, the Jonk River Basin reveals K values ranging from 0.131 to 0.139 t ha h MJ^{-1} mm^{-1} (Fig. 4). Although the variation is narrow, areas in the central and northern basin exhibit slightly higher erodibility, indicating soils with finer textures or weaker structure. Such soils are less cohesive and more prone to detachment when exposed to erosive rainfall. In contrast, areas with coarser or well-structured soils demonstrate slightly lower erodibility values. Despite this, these differences in K values play a significant role when they interact with slope and rainfall erosivity factors.

4.3 Slope characteristics and topography factor (LS factor)

Topography plays a dominant role in shaping erosion patterns. The basin's slope distribution ranges from nearly level terrain (0–1%), mainly in the central and southern plains, to very steep slopes (> 50%), predominantly in the northern and northeastern highlands. Gentle slopes tend to retain moisture and support deposition, whereas steep and very steep slopes accelerate runoff, leading to higher soil detachment and transport rates. Areas with moderate to steep gradients (8–30%) represent zones of particularly high erosion risk due to the combination of sufficient slope and erodible soil cover (Fig. 3).

The slope length and steepness (LS) factor is derived from the SRTM DEM and represents the influence of terrain configuration on soil erosion. LS values in the basin range from 0.00 to 50.69, with the highest values concentrated in hilly, undulating, and escarpment areas where both slope length and gradient amplify runoff velocity. Steeper slopes enhance the gravitational force acting on water flow, while longer slopes extend the accumulation of runoff, leading to greater erosive energy [102],Fig. 4). Valley regions, ridge lines, and steep escarpments thus emerge as critical erosion-prone zones in the LS factor map. Conversely, flat and gently sloping agricultural lands show lower LS values, indicating lesser topographic influence on erosion.

4.4 Vegetation cover (NDVI) and cover management (C-factor)

Normalized Difference Vegetation Index (NDVI) values range from −0.178841 (low vegetation/barren land) to 0.441068 (dense vegetation/forest cover). High NDVI values

correspond to dense forest and well-vegetated croplands, providing canopy cover and reducing soil exposure to rainfall. Conversely, low NDVI areas, including sandbars, degraded croplands, and barren lands, indicate weak or no vegetation cover, leaving soils highly exposed to erosional forces. The distribution shows that northern and northeastern parts of the basin, as well as certain scattered patches in the central and southern zones, record higher NDVI values, signifying the presence of dense forest and croplands with good vegetation cover. In contrast, lower NDVI values are predominantly observed in the eastern and southern margins, where barren lands, built-up areas, and sandy patches dominate, reflecting stressed vegetation (Fig. 3).

The cover-management (C) factor captures the effect of vegetation cover and land management practices on soil erosion. C values in the basin range between 0.076 (dense vegetation) and 0.575 (sparsely vegetated or bare areas). Forested regions and areas with dense vegetation act as protective covers by reducing raindrop impact, intercepting rainfall, enhancing infiltration, and binding the soil with root systems. In contrast, croplands, barren soils, and open scrublands exhibit high C values, reflecting limited ground cover and a higher propensity for erosion. Seasonal cropping cycles further modulate the C factor, as post-harvest agricultural fields are left exposed, significantly increasing erosion risks during pre-monsoon and monsoon periods (Fig. 4).

4.5 Land use/land cover and conservation practice factor (P-factor)

The LULC map is classified into land use categories such as cropland, dense forest, open forest, barren land, built-up areas, sandbars, and water bodies. Cropland is the dominant land use in the study area, particularly in the lower basin, where seasonal agriculture exposes soils to erosion during ploughing and post-harvest periods. Dense forests, concentrated in the northern uplands, act as natural barriers against soil erosion by reducing raindrop impact, promoting infiltration, and stabilizing slopes. Open forests and barren lands, scattered across the basin, represent transitional zones where sparse vegetation increases vulnerability to surface runoff and soil loss. Built-up areas and sandbars further modify surface runoff pathways (Fig. 3). These LULC categories are used to derive the C-factor, which represents the protective influence of vegetation cover and land management practices in the erosion model.

The conservation practice (P) factor accounts for the presence or absence of soil and water conservation measures such as contour farming, strip cropping, terracing, and bunding. In this study, P values were assigned by integrating land use/land cover classes with slope categories following established guidelines [140]. The values range from 0 (effective conservation practices) to 1 (no conservation measures). Most agricultural and open land areas in the Jonk Basin exhibit higher P values, indicating a lack of effective conservation interventions. This is particularly critical in regions with steep slopes and high rainfall erosivity, where inadequate management practices amplify soil erosion intensity. Lower P values are primarily associated with vegetated or naturally protected areas, where the landscape inherently reduces runoff (Fig. 4).

4.6 Soil erosion hotspots and soil loss across sub-basins

The estimated RUSLE model results (Fig. 5) show the spatial distribution of the annual soil loss rate of the study basin and its sub-basins. Table 3 reveals that 85.75% (2961.91 km^2) of the total basin area experiences annual soil loss at a rate of less than

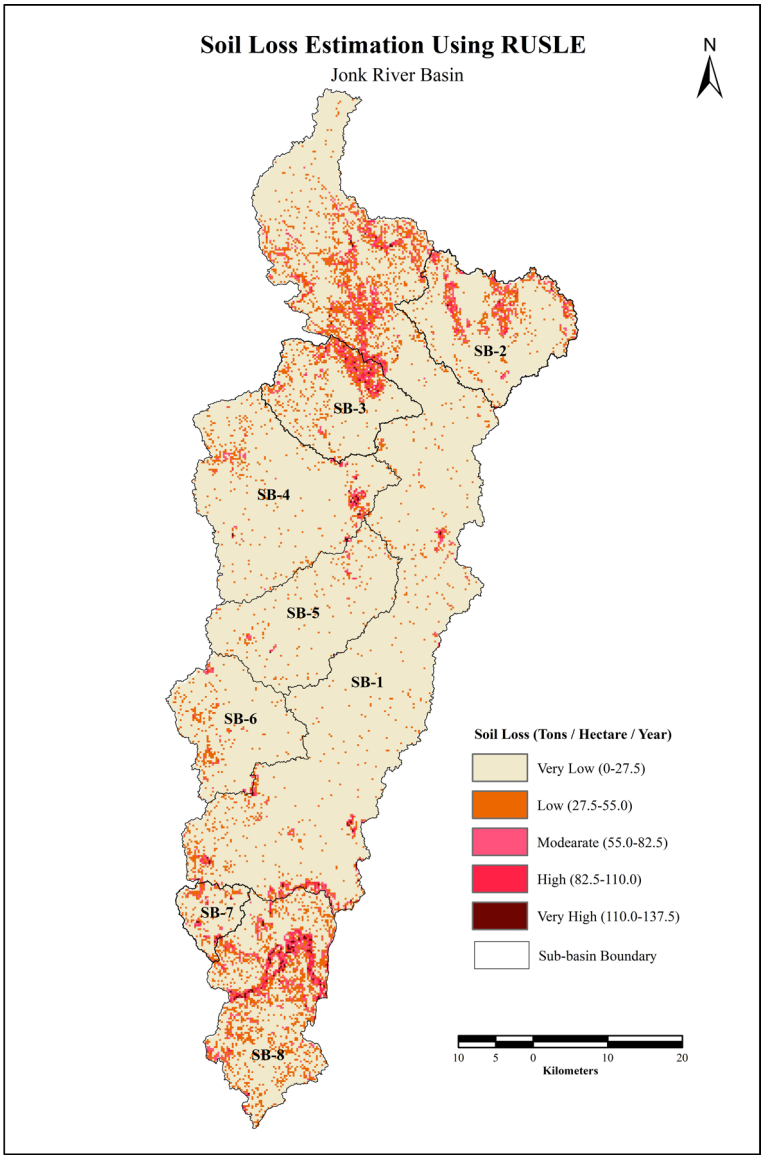


Fig. 5 Soil erosion rate (RUSLE) in the study area. *Source*: Prepared using NASA power rainfall data, FAO soil, NDVI and LULC layers based on Landsat 9 imagery 2022, and slope layer based on SRTM DEM

100 t ha^{-1} yr^{-1}, and this is well distributed across the basin. A total of 9.44% of the basin area experiences soil loss at rates between 27.5 and 55.0 t ha^{-1} yr^{-1}. Areas of moderate (55.0–82.5 t ha^{-1} yr^{-1}), high (82.5–110.0 t ha^{-1} yr^{-1}), and very high (>110.0 t ha^{-1} yr^{-1}) annual soil loss are distributed over 2.89%, 1.45%, and 0.47% of the basin area, respectively. Areas of moderately high and very high soil erosion rates are mainly located along the slopes of hills and plateaus, concentrated in the southern, northern, and northeastern parts of the basin (Fig. 5). Table 3 also presents the sub-basin-wise areal distribution of annual soil loss. From the table, it is observed that 85–95% of the area in different sub-basins experiences soil loss of less than 27.5 t ha^{-1} yr^{-1}.

Among all sub-basins, SB-8 records the highest soil loss, accounting for 21.93%, 6.55%, 3.84%, and 1.17% of the total basin area under the 27.5–55.0, 55.0–82.5, 82.5–110.0, and >110.0 t ha^{-1} yr^{-1} soil loss categories, respectively. SB-8 is the most affected sub-basin because it is characterized by moderately to highly dissected plateau terrain with hills

Table 3 Sub-basin wise area (km^2) under various soil erosion rate (t ha^{-1} yr^{-1}) categories

Sub-basins	Area (km^2) under soil erosion rate (t ha^{-1} yr^{-1}) categories					Total area (km^2)
	Very low	Low	Moderate	High	Very high	
SB-1	1362.22 (88.07%)	124.45 (8.05%)	38.17 (2.47%)	16.85 (1.09%)	5.05 (0.33%)	1546.73 (100%)
SB-2	234.89 (81.92%)	31.91 (11.13%)	13.65 (4.76%)	5.47 (1.91%)	0.81 (0.28%)	286.73 (100%)
SB-3	169.81 (78.03%)	30.76 (14.13%)	10.43 (4.79%)	5.82 (2.67%)	0.81 (0.37%)	217.62 (100%)
SB-4	415.34 (93.20%)	22.56 (5.06%)	4.19 (0.94%)	2.63 (0.59%)	0.91 (0.20%)	445.63 (100%)
SB-5	293.00 (96.75%)	7.70 (2.54%)	1.50 (0.49%)	0.46 (0.15%)	0.17 (0.06%)	302.83 (100%)
SB-6	201.92 (91.47%)	14.55 (6.59%)	2.45 (1.11%)	1.12 (0.51%)	0.72 (0.33%)	220.76 (100%)
SB-7	49.96 (78.79%)	8.95 (14.11%)	2.96 (4.67%)	0.94 (1.49%)	0.60 (0.94%)	63.41 (100%)
SB-8	246.36 (66.51%)	81.22 (21.93%)	24.25 (6.55%)	14.23 (3.84%)	4.32 (1.17%)	370.38 (100%)
Jonk basin	2961.91 (85.75%)	326.02 (9.44%)	99.71 (2.89%)	50.12 (1.45%)	16.31 (0.47%)	3454.08 (100%)

Source: Computed based on Fig. 5

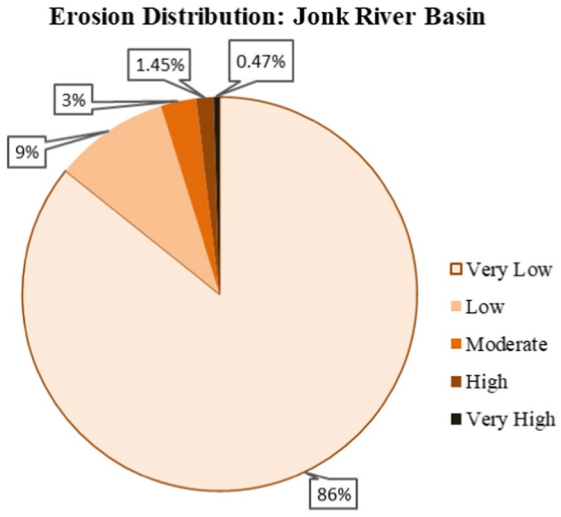


Fig. 6 Percentage distribution of soil erosion severity classes (very low, low, moderate, high, and very high) in the Jonk River Basin derived from the RUSLE model. *Source*: Prepared using Table 3

and valleys. Similarly, sub-basin SB-5 exhibits the lowest soil loss rates, with 96.75% of its total area experiencing soil loss of less than 27.5 t ha^{-1} yr^{-1}, while only 3.25% of the area falls within higher soil loss categories (27.5–55.0, 55.0–82.5, 82.5–110.0, and >110.0 t ha^{-1} yr^{-1}).

The distribution of soil erosion intensity across the Jonk River Basin (Fig. 6) shows that 86% of the basin area falls under the very low erosion class, indicating that most parts of the basin experience minimal soil loss. This suggests relatively stable land surfaces with moderate slopes, vegetation cover, and limited erosive forces across large portions of the basin. The low erosion category accounts for about 9% of the total basin area, representing regions where soil loss occurs at a slightly higher rate but still remains

within manageable limits. These areas may correspond to agricultural lands or gently sloping terrains where rainfall runoff contributes to moderate soil displacement. The moderate erosion class covers approximately 3% of the basin area. These areas generally occur in zones with steeper slopes, reduced vegetation cover, or more intense surface runoff, making them more vulnerable to soil degradation. Higher erosion intensity classes occupy comparatively smaller proportions of the basin. The high erosion category represents about 1.45%, while the very high erosion category constitutes only 0.47% of the total area. These zones are primarily concentrated in the upper reaches and steep terrains of the basin, where slope gradients, sparse vegetation, and intense monsoonal rainfall accelerate soil detachment and transport. Overall, the analysis indicates that although the majority of the Jonk River Basin experiences low levels of soil erosion, certain localized areas remain highly vulnerable and require targeted soil conservation measures to prevent further land degradation.

Figure 7 presents a comparison of the spatial distribution of high and very high soil erosion zones across different sub-basins of the Jonk River Basin. The results indicate considerable variation in erosion intensity among the sub-basins. Among all sub-basins, SB-8 exhibits the highest proportion of severe erosion, with nearly 5% of its area falling under the high erosion category, making it the most erosion-prone sub-basin. This may be attributed to steeper slopes, fragile soil conditions, and diminishing vegetation cover in the upper reaches of the basin. SB-3 also shows relatively high erosion intensity, with approximately 3% of its area under high erosion, suggesting significant vulnerability to soil degradation. Similarly, SB-7 records about 2.4% of its area under high erosion, indicating moderate susceptibility compared with other sub-basins. In contrast, SB-1 and SB-2 are mainly characterized by moderate erosion zones, covering around 1.4% and 2.2% of the area, respectively. These areas likely experience moderate runoff and land surface disturbance, which contribute to soil loss but at lower intensity than high-risk zones. The remaining sub-basins (SB-4, SB-5, and SB-6) show relatively lower erosion intensity, with most areas falling under low or moderate erosion categories and only minimal high-erosion coverage. Notably, SB-5 exhibits the lowest erosion susceptibility, indicating relatively stable terrain and better land cover conditions.

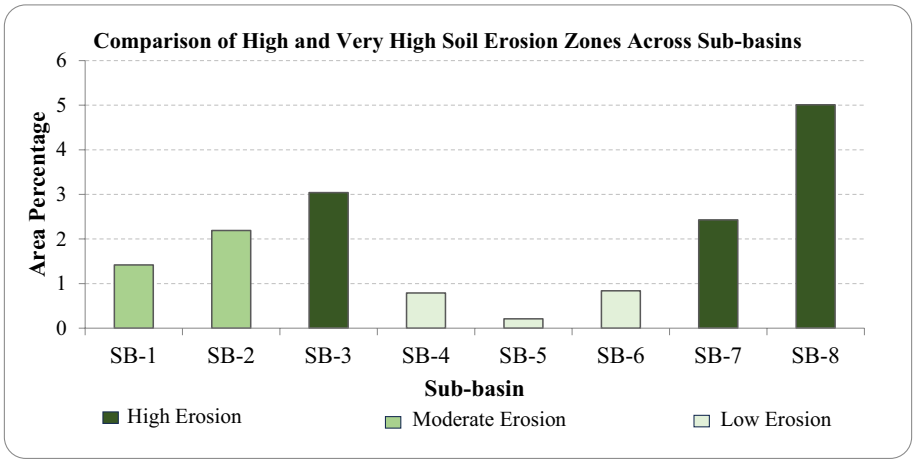


Fig. 7 Comparison of the percentage area under low, moderate, and high soil erosion classes across the sub-basins of the Jonk River Basin. *Source*: Prepared using Table 3

Table 4 Descriptive statistics, Pearson correlation coefficients, and Random Forest-based relative importance of RUSLE factors in the Jonk river basin

Factors	Mean	Min	Max	Standard deviation	Pearson correlation coefficient (r)	RF importance (%IncMSE)	Normalised importance (%)
R	697.86	643.90	733.50	21.88	−0.06	14.24	12.99
K	0.14	0.13	0.14	0.004	0.10	19.30	17.60
LS	0.94	0.00022	14.78	1.05	0.70	34.39	31.37
C	0.27	0.17	0.47	0.025	−0.09	14.22	12.97
P	0.12	0.037	0.58	0.04	0.49	27.48	25.06

Source: Prepared using RUSLE input layers in R environment

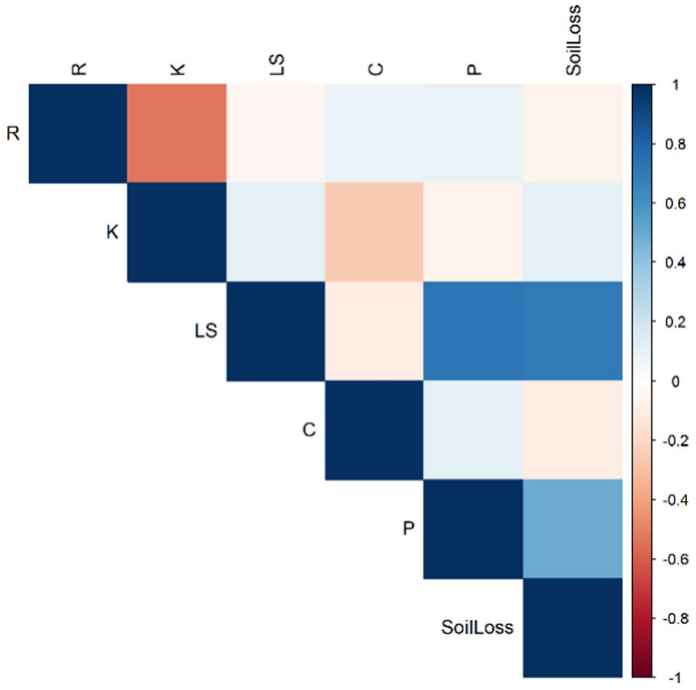


Fig. 8 Correlation heat map showing the relationships among the major soil erosion controlling factors. *Source*: Prepared using RUSLE input layers

4.7 Statistical association and relative importance of the factors on soil loss

4.7.1 Descriptive statistics and Pearson correlation of RUSLE factors with soil loss

The statistical summary (Table 4) of the RUSLE factors provides insight into their distribution and relative contribution to soil erosion within the Jonk River Basin. The rainfall erosivity factor (R) exhibits the highest mean value (697.86), with values ranging from 643.90 to 733.50 and a standard deviation of 21.88, indicating relatively moderate spatial variability in rainfall erosivity across the basin. Despite its high magnitude, the correlation analysis (Fig. 8) reveals a weak negative relationship with soil loss (−0.065), suggesting that rainfall erosivity alone does not strongly control the spatial distribution of erosion within the basin.

The soil erodibility factor (K) shows a low mean value (0.136) with a narrow range (0.131–0.139) and a very small standard deviation (0.0036), indicating that soil properties are relatively uniform across the basin. Its correlation with soil loss (0.104) suggests a weak positive influence on erosion processes.

The topographic factor (LS), representing slope length and steepness, shows the strongest influence on soil erosion. Although the mean value is relatively low (0.94), it exhibits a very wide range (0.00022–14.78) and the highest standard deviation (1.05), indicating significant topographic variability. The LS factor has the highest importance score (45.43) and the strongest positive correlation with soil loss (0.705). LS exhibited the strongest statistical association with estimated soil erosion among the considered RUSLE factors.

The cover management factor (C) has a mean value of 0.275 with moderate variability (standard deviation = 0.0258). However, its correlation (Fig. 8) with soil loss is weak and negative (–0.090), which suggests that areas with relatively better vegetation cover tend to experience lower erosion rates.

The conservation practice factor (P) shows moderate variability, with values ranging from 0.037 to 0.578 and a standard deviation of 0.040. It has a relatively strong positive correlation (Fig. 8) with soil loss (0.495) and a high importance score (26.67), indicating that the absence or inefficiency of conservation practices significantly contributes to increased erosion in certain areas.

4.7.2 Random Forest-based relative importance of RUSLE factors in soil erosion estimation

The Random Forest-based variable importance analysis (Table 4; Fig. 9) indicated that the LS factor exerted the highest influence on soil erosion variability within the Jonk River Basin, contributing approximately 31.37% to the overall model importance. The comparatively higher contribution of the LS factor highlights the significant role of topographic controls, particularly slope length and steepness, in governing erosion susceptibility across the rugged terrain of the Jonk River Basin. The P factor represented the second most influential variable, contributing 25.07%, suggesting that conservation practices and land management measures substantially affect soil erosion dynamics in the study area.

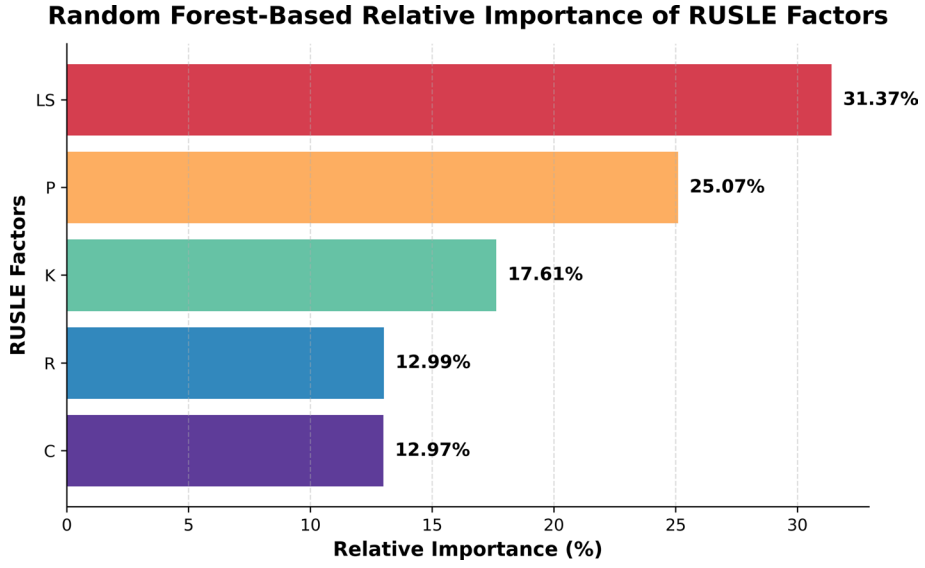


Fig. 9 Relative importance of the RUSLE factors for soil erosion assessment of Jonk river basin: rainfall erosivity (R), soil erodibility (K), slope length–steepness (LS), cover-management (C), and support practice (P). *Source*: Prepared using Table 4

The K factor accounted for 17.61% of the total importance, indicating a moderate influence of soil erodibility on erosion susceptibility. In contrast, the R and C factors exhibited relatively lower contributions, accounting for 12.99% and 12.97%, respectively. This suggests that although rainfall erosivity and land cover conditions influence erosion processes, their relative contribution to the spatial variability of soil loss within the basin was comparatively lower than topographic and conservation-related factors.

Overall, the Random Forest analysis demonstrates that topographic characteristics and conservation practices play more prominent role in explaining soil erosion variability in the Jonk River Basin compared to other RUSLE parameters.

5 Validation of soil erosion estimates

5.1 Spatial validation with erosion sites

Figure 10 illustrates the spatial distribution of soil erosion risk across the Jonk River Basin, categorized into five erosion severity classes ranging from very low (0–27.5 t ha^{-1} yr^{-1}) to very high (110–137.5 t ha^{-1} yr^{-1}), along with its cross-validation against field-observed erosion sites (Fig. 11). A strong correspondence was observed, with most of the erosion-prone areas coinciding with moderate to very high erosion zones in the model output. Particularly, the southern (SB-8) and central (SB-3, SB-1, and SB-2) sub-basins showed concentrated high-risk erosion zones, which are consistent with field evidence of active gully formation and topsoil loss. The agreement between modelled and observed erosion locations highlights the model's ability to capture spatial variability in erosion susceptibility.

5.2 ROC curve and AUC analysis

For quantitative validation, ROC curve analysis was performed to assess the discrimination capacity of the RUSLE predictions (Fig. 12). The ROC curve compares the true positive rate (sensitivity) against the false positive rate (1 – specificity), providing a comprehensive measure of model accuracy. The Area Under the Curve (AUC) was found to be 0.97, which falls within the excellent prediction category [52]. This high AUC score demonstrates that the RUSLE model, integrated with GIS and remote sensing inputs, is capable of reliably distinguishing between erosion-prone and non-erosion-prone areas.

The dual-validation approach, combining field verification with statistical testing, confirms the effectiveness of the RUSLE framework in predicting soil erosion risk within the Jonk River Basin. The strong spatial correspondence with observed erosion sites, along with the excellent AUC score, indicates that the model results can serve as a scientific basis for watershed management, conservation planning, and erosion mitigation strategies.

6 Comparative analysis of RUSLE-based soil erosion rates across Indian river basins

A comparative analysis of RUSLE-based soil erosion rates across Indian river basins shows wide variability depending on climatic, topographic, and land-use conditions (Table 5). Larger basins, such as the Upper Krishna Basin, exhibit very high average annual soil loss (~ 237 t ha^{-1} yr^{-1}), often driven by extensive agriculture and high rainfall erosivity. In contrast, forested or hill catchments such as Manimala (~2.42 t ha^{-1} yr^{-1})

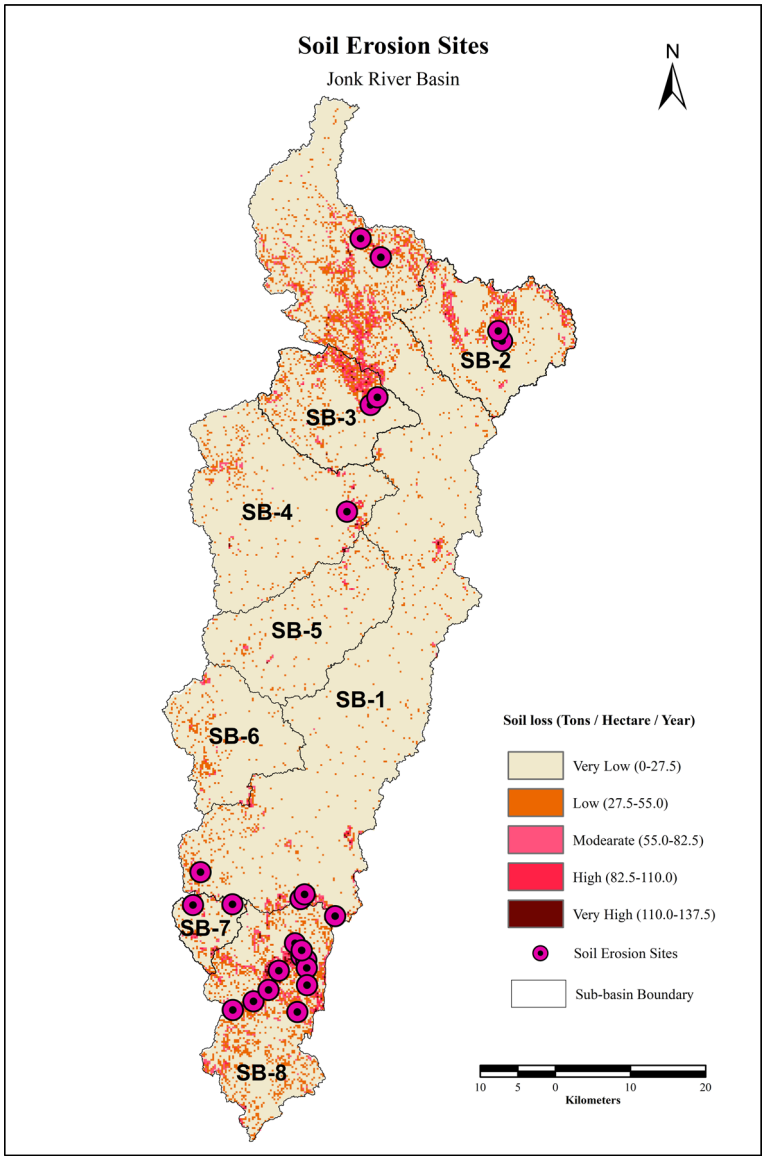


Fig. 10 Soil erosion sites in the study area

and Shambhavi (~0.69 t ha^{-1} yr^{-1}) exhibit much lower average losses due to protective land cover and less intense runoff.

The Mahanadi River Basin, assessed over multiple years, shows mean RUSLE-derived soil loss values in the range of approximately 31.9–37.0 t ha^{-1} yr^{-1}, reflecting mixed land use and river basin characteristics typical of large alluvial systems subject to both cultivation and monsoonal rainfall variability. Sub-catchment studies within the Mahanadi system, such as the Chilika watershed (~ 32.41 t ha^{-1} yr^{-1}), also indicate similar soil loss magnitudes, underscoring the influence of watershed scale and land cover on erosion intensity. Based on NBSS & LUP soil erosion classification, annual soil loss rates are categorized as very slight (<5 t ha^{-1} yr^{-1}), slight (5–10), moderate (10–15), moderately severe (15–20), severe (20–40), very severe (40–80), and extremely severe (> 80) [74, 122]. Therefore, erosion rates reported for the Mahanadi River Basin, which generally

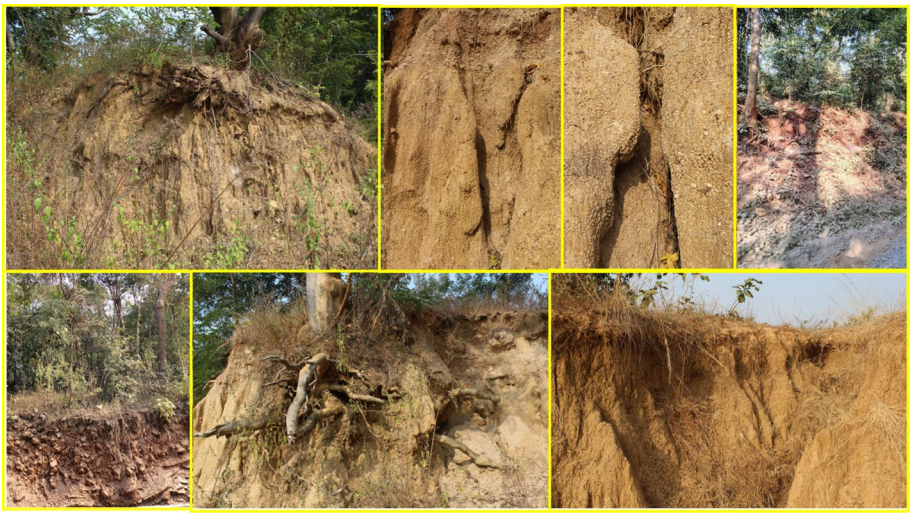


Fig. 11 Exposed slopes and gullies at soil erosion sites in the study area. *Source*: Field images taken during field visit

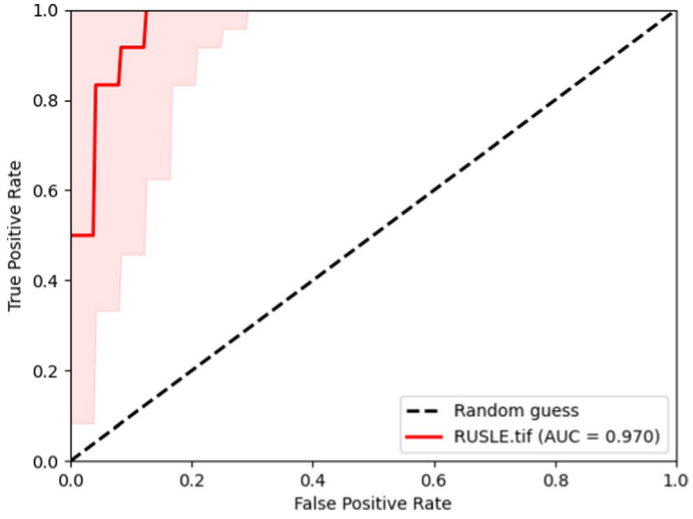


Fig. 12 Validation of RUSLE results using area under ROC curve. *Source*: Prepared using Fig. 5

Table 5 Comparative analysis of RUSLE-based soil erosion rates across Indian river basins

River basin/watershed	Average soil erosion rate (t ha^{-1} yr^{-1})	References/sources
Upper-Krishna River Basin	~237 t ha^{-1} yr^{-1}	Patil and Panhalkar [97]
Silabati River Basin (Eastern India)	~4.35 t ha^{-1} yr^{-1}	Pal and Jana [88]
Muthirapuzha River Basin (western Ghats)	~14.36 t ha^{-1} yr^{-1}	Thomas et al. [130–132]
Karimpuzha River Basin (Western Ghats)	~9.3 t ha^{-1} yr^{-1}	Achu and Thomas [1]
Manimala River Basin (Western Ghats)	~0.04 mm yr^{-1} to 61.80 mm yr^{-1}	Vijith et al. [135]
Thamirabarani River Basin (Southern India)	<0.4–10.8 t ha^{-1} yr^{-1}	Panika et al. [95]
Mahanadi River Basin (average variability)	~31.9–37.0 t ha^{-1} yr^{-1}	Dash and Maity [31]
Chilika watershed (Mahanadi sub-catchment)	~32.41 t ha^{-1} yr^{-1}	Behera et al. [17]



Fig. 13 **A** Harden soil condition, **B**,**C** Murrum soil at upper reaches of Jonk river basin. *Source*: Field images taken during field visit

fall within the moderate to severe categories (30–37 t ha^{-1} yr^{-1}), indicate substantial susceptibility to soil loss under prevailing land-use and monsoonal rainfall conditions.

In comparison, the sub-basin-wise analysis of the Jonk River Basin reveals significant spatial heterogeneity, with sub-basins characterized by dissected plateaus and hilly terrain exhibiting soil loss in the severe to very severe classes, while relatively gentle and well-vegetated sub-basins largely fall within the slight to moderate erosion categories. This contrast highlights the critical influence of topography, land-use patterns, and conservation practices on erosion intensity and underscores the need for prioritized, sub-basin-specific soil conservation interventions within the Mahanadi River system and its tributaries. This comparative assessment is significant as it demonstrates that basin-averaged erosion estimates alone may mask critical intra-basin variability, particularly within large river systems such as the Mahanadi. By explicitly identifying erosion severity at the sub-basin scale, the present study provides a scientific basis for prioritizing soil and water conservation measures where they are most urgently required.

7 Discussions

The RUSLE-based soil erosion assessment for the Jonk River Basin provides valuable insights into the spatial distribution and severity of erosion risk across the basin. While the model highlights priority areas requiring intervention, understanding the broader context of existing land management practices, conservation measures, and agricultural decisions is equally important. The following discussion integrates field observations with regional agricultural dynamics to evaluate current challenges and examines soil conservation strategies for sustainable watershed management.

7.1 Presence of lateritic Murrum soil and its contribution to soil erosion

Murrum soil, a lateritic soil formed from the weathering of basalt, is common in tropical India, particularly across the Deccan Plateau. It is coarse-textured, rich in iron and aluminium oxides, and typically reddish-brown in colour [136]. In the Jonk River basin, it mainly occurs in the elevated, undulating upper reaches (SB-8 and SB-1). Murrum soil is highly erosion-prone during the monsoon. Intense rainfall and sparse vegetation make it loose and susceptible to sheet and rill erosion [63]. In dry periods, it hardens, and reduces infiltration and increases runoff, which further accelerates erosion [117]. Field observations in the upper basin validated these conditions, with visible soil condition, and exposed roots (Figs. 11, 13).

7.2 Cropping practices and soil degradation

Despite the potential of climate-resilient agriculture, field observations indicate that monocropping of paddy and cotton is widely practiced throughout the basin. Such monocropping depletes specific soil nutrients without allowing time for regeneration, which accelerates soil degradation, reduces organic matter, and weakens the soil's capacity to retain moisture. This problem is particularly severe in drought-prone areas, where rainfall variability intensifies the risks of erosion and land degradation.

7.3 Impact of paddy cultivation

Paddy cultivation further impacts soil health through puddling, the practice of tilling the soil under flooded conditions. While puddling reduces seepage and suppresses weeds, it leads to compaction, formation of a hardpan, and reduced porosity [16, 18, 116]. These effects worsen under alternating droughts and floods. During droughts, compacted soils harden and crack, lowering microbial activity and soil fertility [45]. When heavy rains or flash floods follow, the poor infiltration capacity of compacted soils results in rapid runoff, nutrient loss, and greater soil erosion [55].

7.4 Cross-border cropping influences

Cropping patterns in the basin are also strongly shaped by its location in the transition zone between Odisha and Chhattisgarh. Chhattisgarh, known as the "rice bowl of India," has influenced rice-dominated farming systems in adjoining western Odisha districts such as Nuapada, Bargarh, Kalahandi, and Balangir. Both regions share similar ecological and socio-economic conditions as part of the Eastern Plateau and Hills agro-climatic zone, marked by rainfed agriculture, undulating terrain, and red-lateritic soils [56]. The strong cross-border linkages between Chhattisgarh and Odisha shape both farming practices and resource use. Intensive paddy cultivation in Chhattisgarh affects groundwater and surface water availability downstream in Odisha [60, 108]. Agricultural input supply chains, seed exchanges, and rice procurement mechanisms also spill over, influencing market choices across state boundaries [94]. Farmers often synchronize their cropping calendars to align with labour availability and the use of shared machinery such as harvesters and threshers [54]. Moreover, Odisha's paddy procurement model has been adapted in line with Chhattisgarh's procurement policies, reinforcing similar cropping decisions [47, 109, 111, 125].

7.5 Farmer awareness

Although soil erosion risk is inherently high in the Jonk Basin due to fragile lateritic soils, steep slopes, and heavy monsoonal rainfall, awareness among farmers regarding long-term erosion vulnerability is less. While they recognize immediate yield fluctuations, their understanding of broader declining soil moisture retention is weak due to limited extension services and lack of access to agro-climatic knowledge [109]. Therefore, enhancing awareness through targeted agricultural extension programs, climate-resilient advisory services, and community-level soil conservation training is critical for promoting sustainable farming systems in the region.

7.6 Suggested measures for soil conservation in the basin

7.6.1 Vegetative measures

Geotextiles (Jute) and coir logs (coconut fibre roll) are effective soil erosion control measures, especially for steep slopes, exposed areas, and regions with Murrum soil. Grasses with dense, fibrous root systems such as *Vetiver*, *Napier*, and *Bermuda grass* contribute significantly to soil binding, reduce the velocity of surface runoff, and prevent the onset of sheet and rill erosion. Their extensive root networks hold the topsoil firmly in place, enhance soil moisture retention, and support the development of vegetative cover essential for erosion control on slopes and degraded lands. *Use of crop residues or organic mulch* to cover exposed soil to reduce direct impact from raindrops is also suggested.

7.6.2 Water conservation structures

Techniques such as *check dams*, *farm ponds*, *percolation tanks*, *contour bunding*, and *rooftop rainwater harvesting* can effectively conserve water at the micro-watershed level [40]. These structures not only reduce the velocity of surface runoff and prevent soil erosion but also enhance infiltration, improve soil moisture, and contribute to aquifer recharge. Field observations in the basin revealed that several check dams have already been constructed, demonstrating their importance for local water management.

7.6.3 Unsustainable monocropping and the need for drought-resilient cropping strategies

The analysis of land use practices in the study area reveals a dominant trend of paddy cultivation, often practiced as monocropping. While paddy is a water-intensive crop, its extensive cultivation under rainfed and drought-prone conditions makes the agricultural system highly vulnerable to climatic variability. Monocropping further exacerbates this vulnerability by reducing soil resilience, depleting nutrients, and increasing susceptibility to both drought stress and soil erosion. Similar concerns regarding the sustainability of rice-based systems under water stress have been highlighted by International Rice Research Institute [23]. Paddy cultivation under limited water availability leads to inefficient water use and declining productivity, particularly during years of below-normal rainfall. Continuous monocropping disrupts soil structure, reduces organic matter content, and weakens the soil's capacity to retain moisture. This not only intensifies drought impacts but also accelerates runoff and soil erosion during high-intensity rainfall events [64].

To address these challenges, a shift towards drought-tolerant and diversified cropping systems is strongly recommended. Crops such as millets (e.g., sorghum and pearl millet), pulses (e.g., pigeon pea and chickpea), and oilseeds (e.g., mustard and groundnut) are well adapted to low-moisture environments and require significantly less water compared to paddy. These crops also improve soil fertility and structure, particularly through biological nitrogen fixation in legumes [107]. In addition to crop diversification, the adoption of efficient irrigation practices is crucial. Techniques such as drip irrigation, sprinkler systems, and alternate wetting and drying (AWD) in rice cultivation have been shown to reduce water use without significant yield loss [23]. Water conservation structures such as farm ponds, check dams, and rainwater harvesting systems further enhance water availability during dry periods [107].

Furthermore, integrating soil and water conservation measures, such as contour bunding, mulching, and agroforestry, can significantly improve soil moisture retention and

reduce runoff and erosion. These practices enhance ecosystem resilience and contribute to sustainable land management [62]. Overall, transitioning from water-intensive monocropping to diversified, drought-resilient cropping systems, combined with improved irrigation and conservation practices, is essential for enhancing long-term agricultural sustainability in the study area.

7.6.4 Protective measures on exposed slopes

The field observations and spatial analysis clearly indicate the presence of several exposed slopes within the study area (Fig. 11). These slopes are highly vulnerable to erosion processes due to the absence of structural stabilization measures such as gabion wall protection and slope netting. In geomorphologically sensitive regions, exposed slopes without reinforcement are prone to accelerated soil loss, particularly under intense rainfall and runoff conditions.

Slope instability is widely recognized as a major geotechnical hazard, often triggered by rainfall infiltration, erosion, and changes in groundwater conditions. If appropriate protective measures are not implemented, such conditions may lead to landslides, soil displacement, and infrastructure damage [67]. The lack of gabion structures in the study area reduces the resistance of slopes against surface runoff and mass movement processes, thereby increasing the likelihood of rill and gully formation. Gabion walls are considered an effective and widely used remedial measure for slope stabilization, as they enhance structural stability and control erosion by dissipating flow energy and providing mechanical support [100].

Similarly, the absence of protective netting or bioengineering techniques fails to provide adequate surface binding, making soil particles more susceptible to detachment and transport. This condition is further aggravated in areas with steep gradients and sparse vegetation cover, where natural resistance to erosion is inherently low. Studies have shown that reinforced systems such as gabion-based or geosynthetic structures can significantly improve slope stability, increase the factor of safety and reduce sediment loss [67].

Moreover, unprotected slopes not only contribute to on-site soil degradation but also increase sediment yield to downstream environments, potentially affecting river morphology and agricultural productivity. The observed condition highlights a significant gap in slope management practices within the region.

Therefore, the implementation of appropriate soil conservation measures, including gabion walls, geo-textile netting, and vegetative stabilization, is strongly recommended to minimize erosion risk and enhance slope stability. Integrating such structural and bioengineering approaches within a watershed management framework can substantially reduce land degradation and improve long-term environmental sustainability.

8 Conclusions

The present study provides a comprehensive spatial assessment of soil erosion in the Jonk River Basin, Easter India, by integrating the Revised Universal Soil Loss Equation (RUSLE) with advanced geospatial techniques. The methodological framework combines GIS-based spatial modelling with statistical computation in the R environment, enabling efficient data processing and validation. By incorporating the five key erosion-controlling factors: rainfall erosivity (R), soil erodibility (K), slope length and steepness

(LS), cover management (C), and conservation practices (P), the study attempts to efficiently delineate erosion-prone zones and capture the spatial heterogeneity of soil loss across this drought-prone basin. The strength of the methodology is further validated through field validation and ROC curve analysis, which yielded a high AUC value of 0.97, demonstrating strong predictive performance.

A major contribution of this research lies in addressing a critical knowledge gap. Previous studies in the Jonk River Basin have primarily focused on groundwater potential mapping, morphometric analysis, and watershed-scale hydrological assessments. While soil erosion assessments exist for the larger Mahanadi River Basin, focused quantitative evaluations of soil erosion dynamics in the Jonk River Catchment have received comparatively little attention, despite the catchment being categorized under severe to very severe soil erosion susceptibility within the Mahanadi River system. This study, therefore, fills an important regional and scientific gap by providing the integrated erosion risk evaluation for the basin. The novelty of the study is further strengthened by its methodological and analytical advancements. The integration of GIS with the R programming environment enables reproducible and scalable analysis, particularly in computing erosion parameters, performing statistical validation, and conducting relative influence analysis. The integration of RUSLE with geospatial techniques provides a strong and scalable framework for erosion risk assessment in semi-arid and drought-prone basins.

The results Random Forest based relative importance analysis reveal that the slope length and steepness (LS) factor and conservation practice (P) factor are the dominant drivers of soil erosion in the basin. The high influence (31.37%) of the LS factor reflects the strong geomorphic control exerted by the dissected plateau terrain, while the prominence of the P factor highlights the critical role of land management and conservation interventions in the basin. This combined evaluation of biophysical and management factors represents a significant advancement in such integrative analysis. The findings indicate considerable spatial variability in erosion risk across sub-basins. Sub-basin SB-8 emerges as the most vulnerable due to its rugged topography and runoff-sediment transport dynamics, whereas sub-basin SB-5 shows relatively low erosion intensity, with the majority of its area experiencing minimal soil loss. Majority of the basin area exhibits relatively low soil erosion rates; however, specific localized zones remain highly susceptible and necessitate focused soil conservation interventions to mitigate further land degradation. Field observations further support these results, revealing both the presence of conservation structures such as check dams and significant gaps, including unprotected exposed slopes and the prevalence of water-intensive paddy monocropping in drought-prone areas. Therefore, the study contributes to sustainable land management by identifying priority zones for intervention. Highly vulnerable sub-basins such as SB-8 and SB-1 require immediate implementation of structural measures, including contour bunding, terracing, gully plugging, and slope stabilization techniques. Additionally, the study highlights the need for transitioning towards drought-resilient agricultural practices, including crop diversification and efficient irrigation methods, to reduce both erosion and water stress. In contrast, relatively stable areas such as SB-5 should focus on maintaining existing conservation practices to sustain soil health and productivity. The study gives actionable insights for policymakers, watershed managers, and local stakeholders. Continuous monitoring and adaptive management using such integrated approaches are

essential for ensuring long-term sustainability of land and water resources in the Jonk River Basin and similar climatic environments.

9 Limitations of the study

- The major limitation of this study is the lack of data availability for the validation of the soil erosion rates estimated using the RUSLE model. Although the Jonk River Basin holds a GDSQ (Gauge and Discharge Site) station located at Rampur, Chhattisgarh, India, sediment discharge data are not available. As a result, the estimated erosion rates could not be compared with observed sediment yield for calibration or accuracy assessment. This restricts the ability to quantify the uncertainty of the model outputs. However, the spatial distribution of soil erosion risk zones was validated through field verification of observed erosion sites, which supports the reliability of the model's spatial predictions despite the lack of sediment data for numerical validation. Therefore, the results presented should be interpreted as indicative estimates of potential soil erosion risk rather than absolute values.
- The integration of high-resolution (30 m) SRTM DEM with comparatively coarse-resolution FAO soil data (~ 1 km) for K-factor estimation may introduce uncertainties in spatial representation. Although the soil data were resampled using the bilinear interpolation method to match the DEM resolution, this downscaling does not fully capture sub-grid variability as this process does not introduce new soil variability but redistributes existing values and may affect the precision of soil erosion estimates at the sub-basin scale.
- One of the key limitations of the present study is related to the availability and consistency of ground-based rainfall data. Although an IMD rain gauge station is located at Baghbahera (Chhatisgarh) within the study region, the observed rainfall data were not continuous and contained significant missing values over the study period. Due to these data gaps, a comprehensive and statistically robust comparison between IMD observations and NASA POWER datasets could not be performed. In this context, the NASA POWER dataset was utilized as an alternative source of rainfall data owing to its spatial continuity, long-term temporal coverage, and easy accessibility. Such gridded datasets are particularly useful in data-scarce regions and have been widely applied in hydrological and soil erosion studies. However, it is important to acknowledge that satellite-derived datasets like NASA POWER have inherent limitations. These include relatively coarse spatial resolution, potential underestimation or overestimation of extreme rainfall events, and limited ability to capture localized variability influenced by topography and micro-climatic conditions. Therefore, while NASA POWER data provides a consistent and practical input for regional-scale modelling, its use may introduce uncertainties in representing actual ground conditions.
- The present study has certain limitations associated with field-based validation of the RUSLE model outputs. Field verification within the Jonk River Basin was particularly challenging in steep and highly erosion-prone slope areas, where erosion signatures were visually evident during field observations. However, direct collection of validation points in such terrains involved substantial accessibility and safety constraints. Consequently, the ROC-AUC validation was performed using a limited number of verified erosion locations. Therefore, the obtained ROC-AUC value

should be interpreted as an indicative measure of model performance rather than an absolute measure of predictive certainty.

- Although Random Forest-based importance analysis provides a more detailed estimate of predictor contribution compared to simple correlation analysis, the derived importance values should still be interpreted as indicative measures of relative predictive influence rather than definitive geomorphic causality.
- The P factor was estimated using slope- and land use-based empirical assignments due to the absence of spatially explicit field-scale conservation-practice datasets for the basin. Consequently, local variations in actual support practices such as contour bunding, terracing, and field management measures may not be fully represented, which could introduce uncertainty in the estimated erosion values. Since the P factor exhibited comparatively high relative importance in the Random Forest-based analysis, uncertainties associated with its generalized estimation approach may have a noticeable influence on the spatial interpretation of erosion susceptibility patterns within the basin.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s44378-026-00259-0.

Supplementary Material 1.

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Author contributions
The first draft of the manuscript was written by K. S. She contributed to the original draft preparation, visualization, validation, software implementation, formal analysis, and manuscript editing. C. M. contributed to validation and formal analysis. R. R. contributed to review and supervision of the study. A. S. contributed to review and editing of the manuscript. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability
All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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